XRPL Validator — Daily Engineering Report · 2026-10-22

Period: 2026-10-22 00:00 to 24:00 US Eastern (2026-10-22 04:00 to 2026-10-23 04:00 UTC); ledgers #107,819,435 to #107,841,423. Every count from the ledger and the validator's logs stops at #107,841,423, closed 03:59:51 UTC. Generated 2026-10-23 00:20 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Status

Normal — coverage outside planned maintenance 100.000% (objective 99.9%), every validation agreed with the network, no open engine receipt.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 21,915 of the 21,989 ledgers the network closed (99.663%); every validation matched the network's validated ledger. Unsigned: 74 during planned restarts.
- **Native engine:** 0 engine receipts.
- **Changes:** 1 deployment (cycle 187), 2 commits merged into main.
- **Network:** 2 slow-close periods; the longest, 14:19 UTC–14:20 UTC, closed ledgers up to 18 s apart.
- **Independence program:** 0 of 30 qualifying days (snapshot at 04:21:04 UTC).
- **Amendment fixCleanup3_4_0** holds majority; activation due ~2026-11-04 14:46 UTC.

Service levels

Measure	Value
Ledgers closed by the network	21,989
Validations issued	21,915
Coverage (validations / ledgers closed)	99.663%
Validations matching the network's validated ledger hash	21,915 of 21,915
Unsigned during planned restarts	74
Unsigned outside maintenance	0
Coverage outside planned maintenance (objective 99.9%)	100.000%
Restarts	1
Basis: our own ledger hash	21,869
Basis: our own hash, witnessed by the reference node (quorum short)	46

Measure	Value
External agreement, data.xrpl.org (UTC day 2026-10-22)	0.99648 (74 missed of 21,047)

Restarts:

- 12:09:08 UTC: deployment of cycle 187; validations resumed at #107,827,030 (74 ledgers unsigned between the last validation before the restart and the first after it)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#107,819,435–#107,841,423), by step:

Step	Ledgers with a receipt
Step 1 · native engine	0
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	38
Step 5 · own hash	0
Writer	0

Live counters, snapshot at 04:21:04 UTC (since the last restart, 2026-10-22 12:09 UTC; not cut off at the period's last ledger):

Step	Checked	Result
1 · native engine vs libxrpl	14,707 ledgers, 2,469,971 transactions	14,707 fully matched; results 0, bytes 0, keys 0/0
2 · transaction records	14,718 ledgers, 2,546,076 records	2,546,076 exact; differ 0, tree 0
3 · transaction order	14,718 ledgers (mode: on)	14,718 exact; order 0, results 0, bytes 0, refused 0
4 · ledger data from peers	14,748 ledgers fetched	14,748 matched; differ 0, failed 0
4 · trusted validations	14,789 ledgers judged	14,789 agreed; differ 0, short of quorum 0
5 · our own ledger hash	14,792 ledgers	14,718 own; differ 0, partial 74
Writer	every ledger written	mismatch 0, distrusted 0

"Partial" ledgers are those the engine could not yet build alone while warming up after a restart.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	08:45 UTC	#107,823,847–#107,823,849 (3)	Step 4 · validations	0–1 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)

ID	Time	Ledger(s)	Step	Description	Disposition
R-02	08:45 UTC	#107,823,858–#107,823,892 (35)	Step 4 · validations	0–16 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-10-22-01, INC-01 is INC-2026-10-22-01.

Incidents

No incidents.

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
187	`e83a4f7`	12:09:58 UTC	2,183 passed / 0 failed; port 1,006 / 0	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches	393 matched, 0 mismatched	#62 from 12:36 UTC: in progress

Cycle 187 delivered: `61be1f3` fix(validator): an empty ledger's transaction tree is ours, so 5a can sign it; `e83a4f7` fix(step4): stage A names a PermissionValue as rippled's JSON does

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
11:25 UTC	xrpl-validator	`e83a4f7`	fix(step4): stage A names a PermissionValue as rippled's JSON does
11:25 UTC	xrpl-validator	`61be1f3`	fix(validator): an empty ledger's transaction tree is ours, so 5a can sign it

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(step4): stage A names a PermissionValue as rippled's JSON does** (`e83a4f7`, xrpl-validator) rippled's JSON names a PermissionValue (STUInt32::getJson through Permission's names): a transaction type's code plus one, or a granular permission from 65537 on. The codec gives the number, so stage A's copy of a DelegateSet differed from the reference node's on mainnet #107804444 4A242D3BBE51: ours 28, the reference node's "NFTokenCreateOffer". respell now names it. The definitions file is read once for both respellings.

- **fix(validator): an empty ledger's transaction tree is ours, so 5a can sign it** (`61be1f3`, xrpl-validator) check_ledger returned no tree for a ledger with no transactions, so our own hash borrowed its transactions (and its order, which the native order never judges for an empty set). Under 5a, which signs only a ledger whose every part is ours, an empty ledger could never be signed. On mainnet the network resumed after the 21 Oct stall with six empty ledgers, #107806253-258, and the validator signed none of them, though …

Independence program

Qualifying days: **0 of 30** (snapshot at 04:21:04 UTC). A UTC day qualifies when every validation is of a ledger our own engine built, with our own transaction order and ledger data taken from peers, and coverage is at least 98%.

UTC day 2026-10-22 — tracked #107,815,768–#107,837,737	Value
Validations	21,845 (21,845 own, 0 borrowed)
Coverage of the tracked ledgers	99.43%
Our own transaction order	on
Ledger data from peers	no

Not qualifying: the ledger data came from the reference node, not peers (Step 4).

The validator saves this record every 64 validations, so a day in progress lags the signing log slightly; it is not cut off at the period's last ledger.

Network

Measure	Value
Ledgers closed	21,989 (#107,819,435–#107,841,423)
Mean close interval	3.93 s
Longest close interval	18 s (before #107,828,936)
Closes at least 15 s apart	2
Empty ledgers	0

- Slow closes 14:19:42 UTC–14:20:00 UTC: ledgers #107,828,936, 1 close at least 15 s apart, up to 18 s.
- Slow closes 14:20:04 UTC–14:20:20 UTC: ledgers #107,828,941, 1 close at least 15 s apart, up to 16 s.

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_4_0	2026-10-21 14:46 UTC	2026-11-04 14:46 UTC

Amendments enabled on mainnet: 97.

Open items

- Committed, awaiting merge and deployment: `987f5de` build: xrpl-core is published as halcyon-xrpl-core; read its definitions from the crate.
- Soak #62 (cycle 187) in progress since 2026-10-22 12:36 UTC; due to complete 2026-10-23 12:36 UTC.
- Amendment fixCleanup3_4_0: activation due ~2026-11-04 14:46 UTC.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
10-21 13:28	10-22 11:25	xrpl-validator	`e83a4f7`	fix(step4): stage A names a PermissionValue as rippled's JSON does
10-21 13:26	10-22 11:25	xrpl-validator	`61be1f3`	fix(validator): an empty ledger's transaction tree is ours, so 5a can sign it

Times in UTC. A commit appears on the day it was committed to main or merged into it.